

15 enriched infographic slides compiled into one presentation

Module 1: Foundations of Statistics

- Introduction to Machine Learning with Statistics
- Nature of Statistics: Collect, Organize, Summarize, Analyze, Conclude
- Statistics Foundation: Variables, Data, Dataset, and Random Variables
- Descriptive vs Inferential Statistics
- Population vs Sample in Statistical Studies

Module 2: Types of Data and Measurement

- Types of Data: Qualitative and Quantitative Variables
- Discrete vs Continuous Data
- Four Levels of Measurement: Nominal, Ordinal, Interval, Ratio

Module 3: Data Collection and Sampling

- Data Collection Methods in Statistics
- Sampling Techniques: Random, Systematic, Stratified, Cluster, Convenience
- Common Sampling Errors and Biases
- Sampling Technique Practice: Identify the Sampling Method

Module 4: Study Design and Application

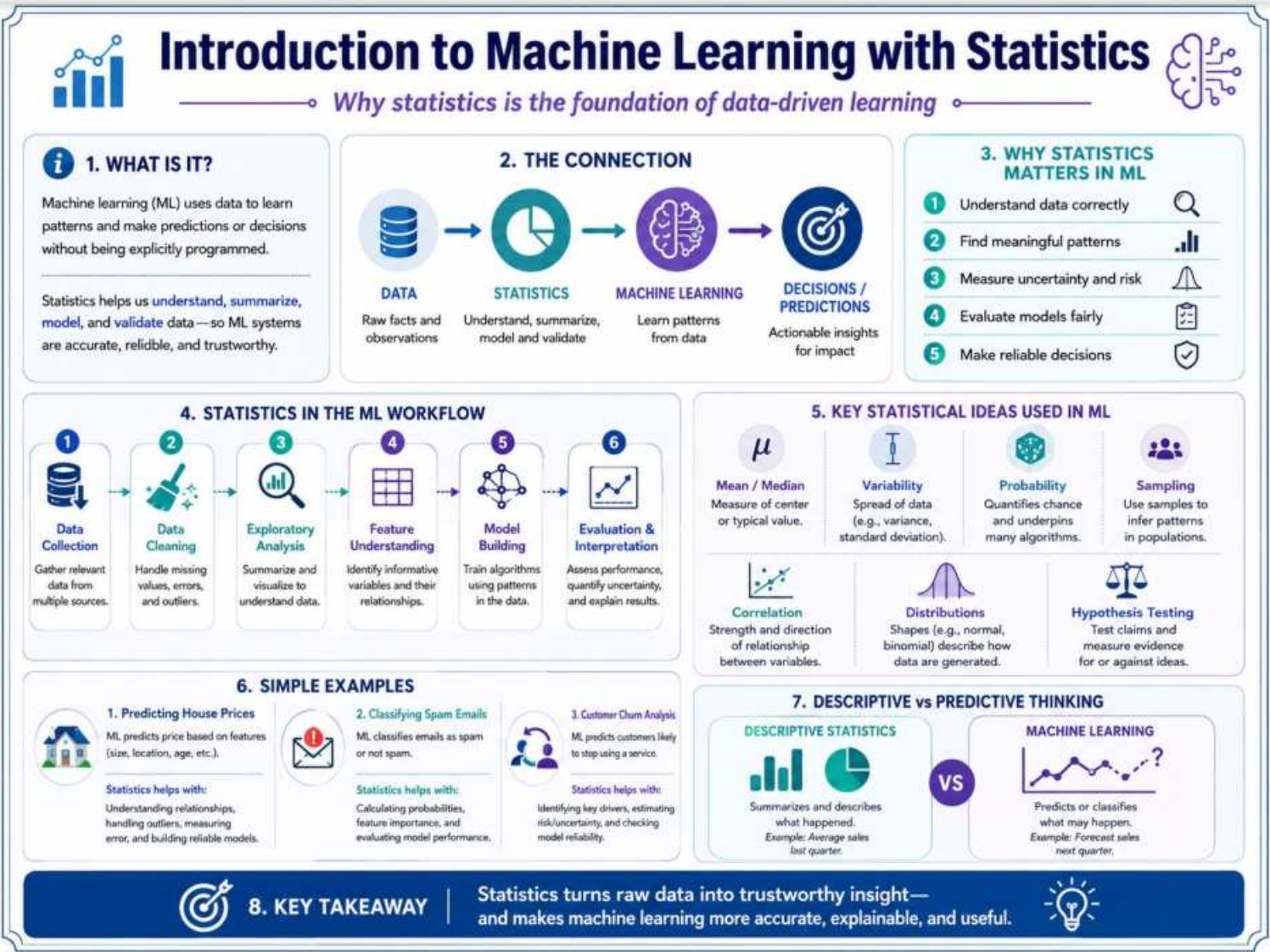
- Observational vs Experimental Studies
- Statistics to Machine Learning Bridge
- Choosing the Right Statistical Method: A Beginner Decision Map

Module 1: Foundations of Statistics

5 infographic slides

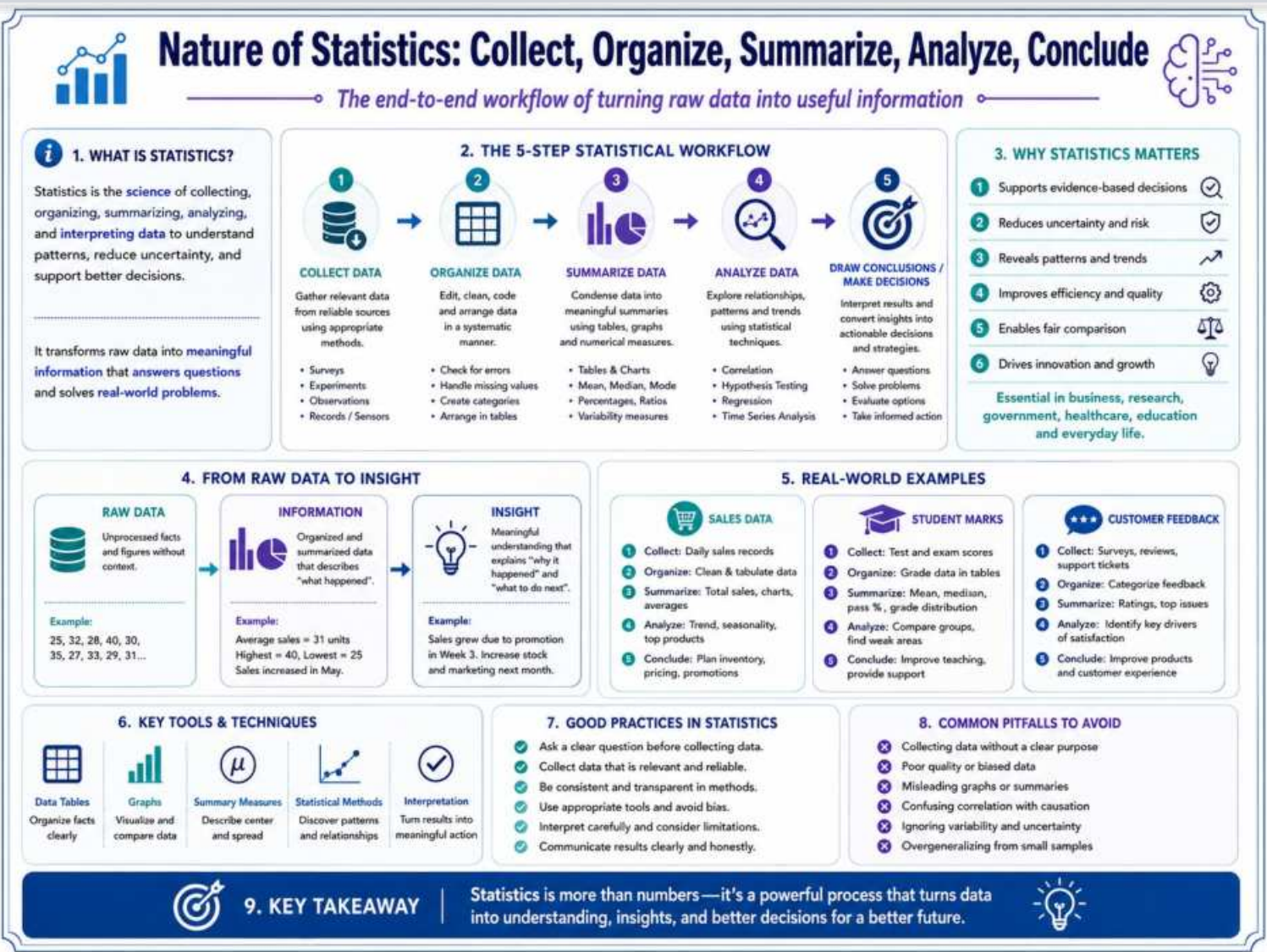
Introduction to Machine Learning with Statistics

Module 1: Foundations of Statistics



Nature of Statistics: Collect, Organize, Summarize, Analyze, Conclude

Module 1: Foundations of Statistics



Statistics Foundation: Variables, Data, Dataset, and Random Variables

Module 1: Foundations of Statistics

Statistics Foundation: Variables, Data, Dataset, and Random Variables

Core building blocks for understanding statistical information

1. KEY TERMS: THE BUILDING BLOCKS

Variable
A characteristic or attribute that can take different values across individuals or objects.
Example: Age, height, gender, income, marks.

Observation (or Record)
Information collected about one individual, object, or unit.
Example: Data about one student or one customer.

Datum (or Data Point)
A single value of a variable for one observation.
Example: Age = 20, Height = 170 cm.

Data
The collection of all data points (values) gathered on variables for all observations.
Example: All ages collected from 100 students.

Dataset
A structured collection of data, usually arranged in rows and columns.
Example: A table of students with age, gender, height.

2. RANDOM VARIABLE EXPLAINED

A random variable is a variable whose value is determined by the outcome of a random process.

Think of it this way:
When we repeat a random experiment (like rolling a die), the result changes. A random variable is the rule that assigns a number to each possible outcome.

Examples:

- X = number shown on a die (values: 1–6)
- Y = number of heads in 3 coin tosses (values: 0–3)
- Z = daily rainfall in mm (can vary from day to day)

3. EXAMPLES OF VARIABLES

Variable	Type	Description	Example Values
Age	Quantitative (Discrete)	Number of years completed.	18, 19, 20, 21, ...
Height	Quantitative (Continuous)	Measurement on a continuous scale.	156.2, 162.5, 170.0, ... (cm)
Gender	Qualitative (Nominal)	Categories with no natural order.	Male, Female, Other
Income	Quantitative (Continuous)	Amount of money earned.	25,000; 45,750; 60,000; ... (\$)
Marks	Quantitative (Discrete)	Scores in an exam (out of 100).	45, 67, 82, 93, 100

4. QUALITATIVE vs QUANTITATIVE VARIABLES

QUALITATIVE (CATEGORICAL)

Describe qualities or categories. Cannot be meaningfully measured with numbers.

Nominal (No order)

Ordinal (Has order)

Examples:
Gender, Color, Blood Group, Satisfaction Level (Low/Medium/High)

QUANTITATIVE (NUMERICAL)

Express counts or measurements. Can be measured and expressed with numbers.

Discrete (Countable)

Continuous (Measurable)

Examples:
Age, Number of siblings, Marks, Height, Weight, Time, Income

5. HOW ROWS AND COLUMNS FORM A DATASET

Columns = Variables
Each column represents a variable (e.g., Age, Height).

Rows = Observations
Each row represents one observation (one individual or unit).

ID	Age	Gender	Height (cm)	Marks
1	...	...	...	...
2	...	...	...	...
3	...	...	...	...
...	...	...	...	...
n	...	...	...	...

A dataset is a table where each row is an observation and each column is a variable.

6. WORKED MINI EXAMPLE

Dataset: Student Information (5 Records)

ID	Age (years)	Gender	Height (cm)	Marks (out of 100)	Income (₹ per month)
1	19	Female	160.5	78	25,000
2	20	Male	172.0	85	30,000
3	18	Female	155.0	92	22,000
4	21	Male	180.3	66	28,000
5	19	Other	168.7	74	26,000

Variables (columns):
Age, Gender, Height, Marks, Income

Observations (rows):
5 students (records 1–5)

7. COMMON MISTAKES STUDENTS MAKE

Confusing variable with value
Variable = the characteristic.
Value = the specific number or category.
Example: 'Age' is a variable; '20' is a value.

Confusing dataset with database
Dataset = a single structured table about one subject or topic.
Database = a collection of multiple related datasets/tables.

Thinking more data is always better
Data quality matters more than quantity. Biased, duplicate, or irrelevant data can mislead analysis.

Treating IDs as meaningful numbers
ID numbers are just labels. Do not use them for calculations (mean, sum, etc.).
Example: ID 1, 2, 3 are not measures.

Ignoring variable type
Using the wrong methods for categorical vs numerical data leads to incorrect conclusions.
Example: Taking average of gender is meaningless.

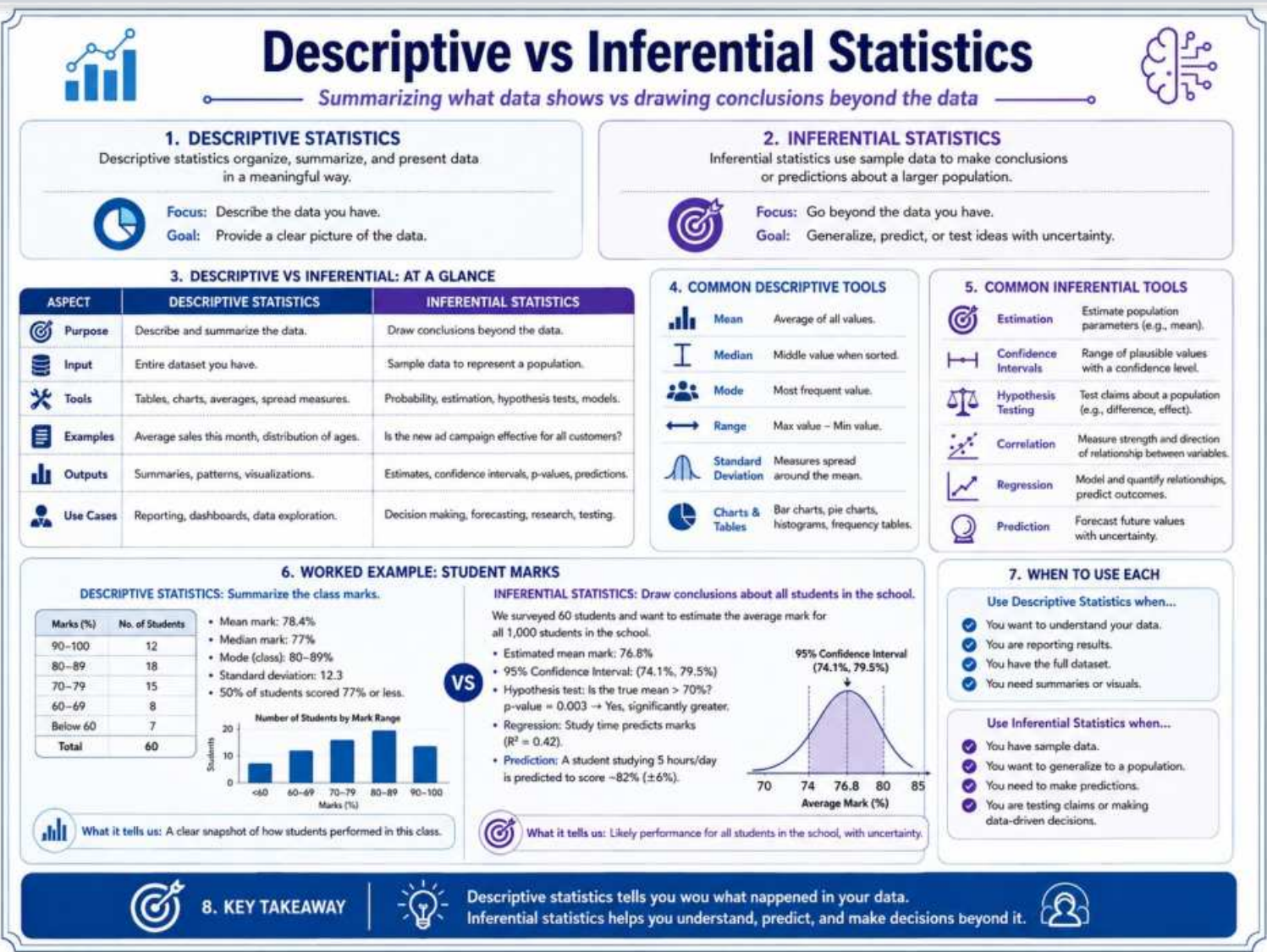
8. KEY TAKEAWAY

Variables describe what we measure. Data are the values we collect. Organized in rows and columns, they form datasets that power statistical insights.

Infographic 3 of 15

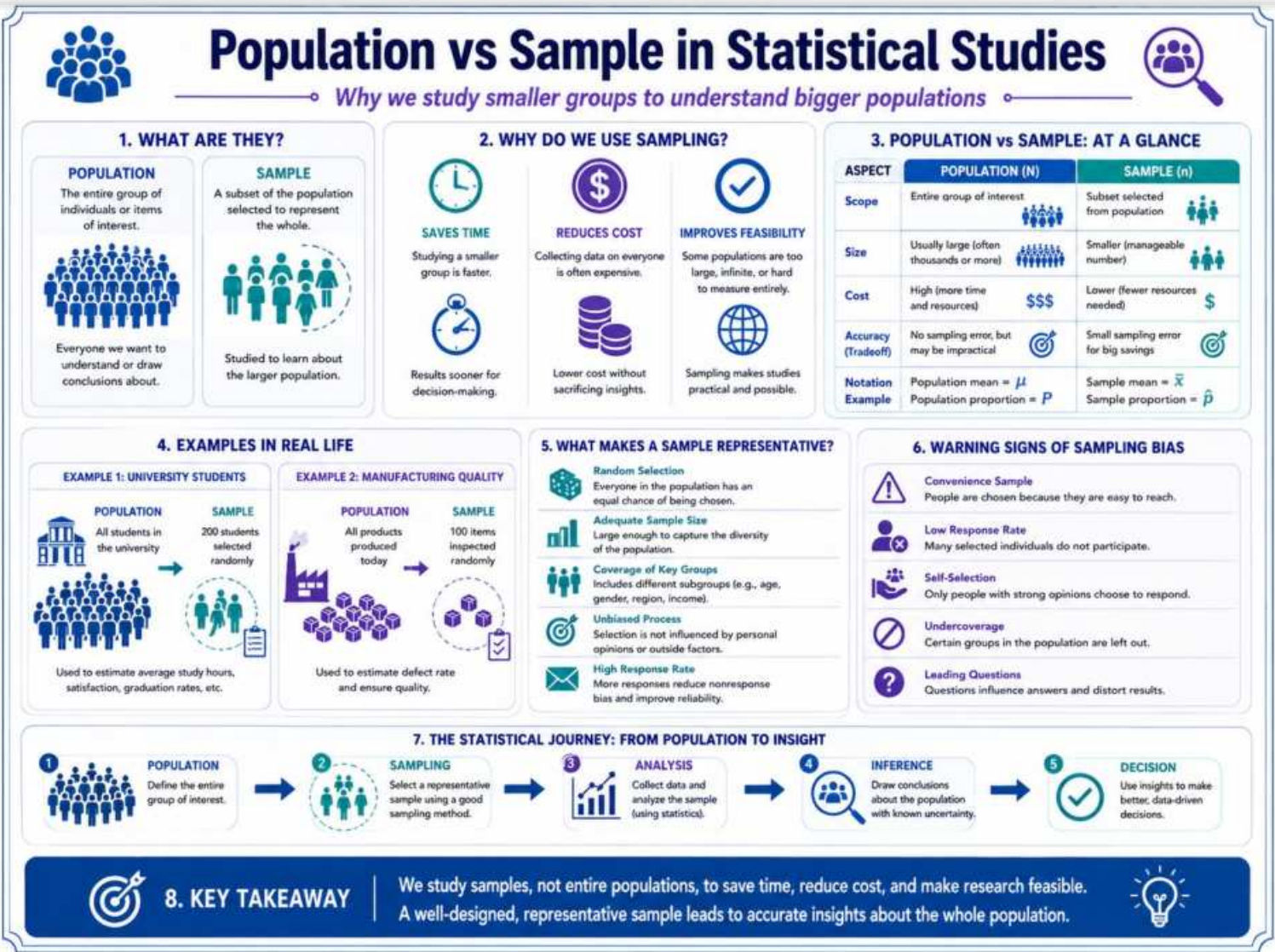
Descriptive vs Inferential Statistics

Module 1: Foundations of Statistics



Population vs Sample in Statistical Studies

Module 1: Foundations of Statistics

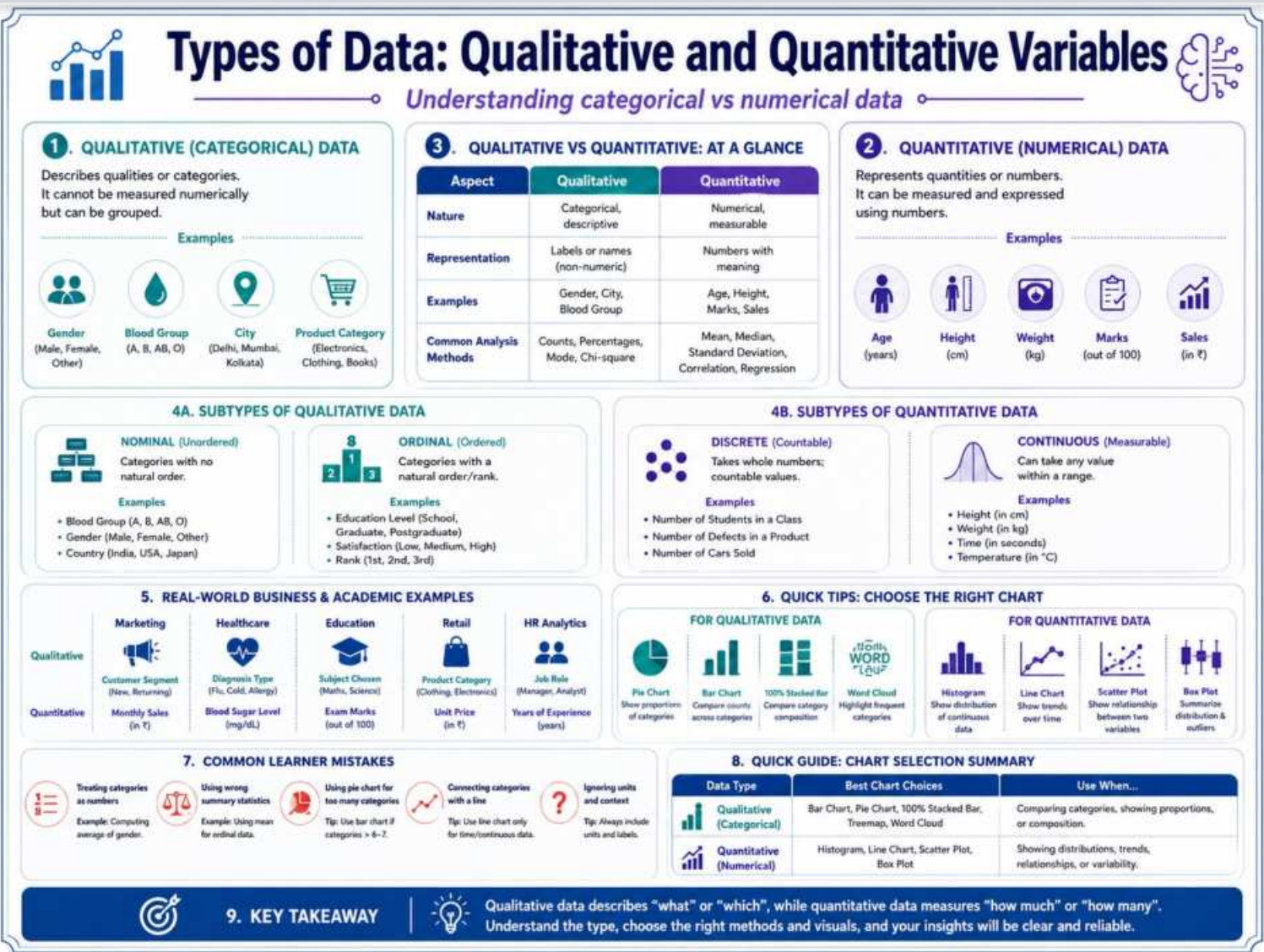


Module 2: Types of Data and Measurement

3 infographic slides

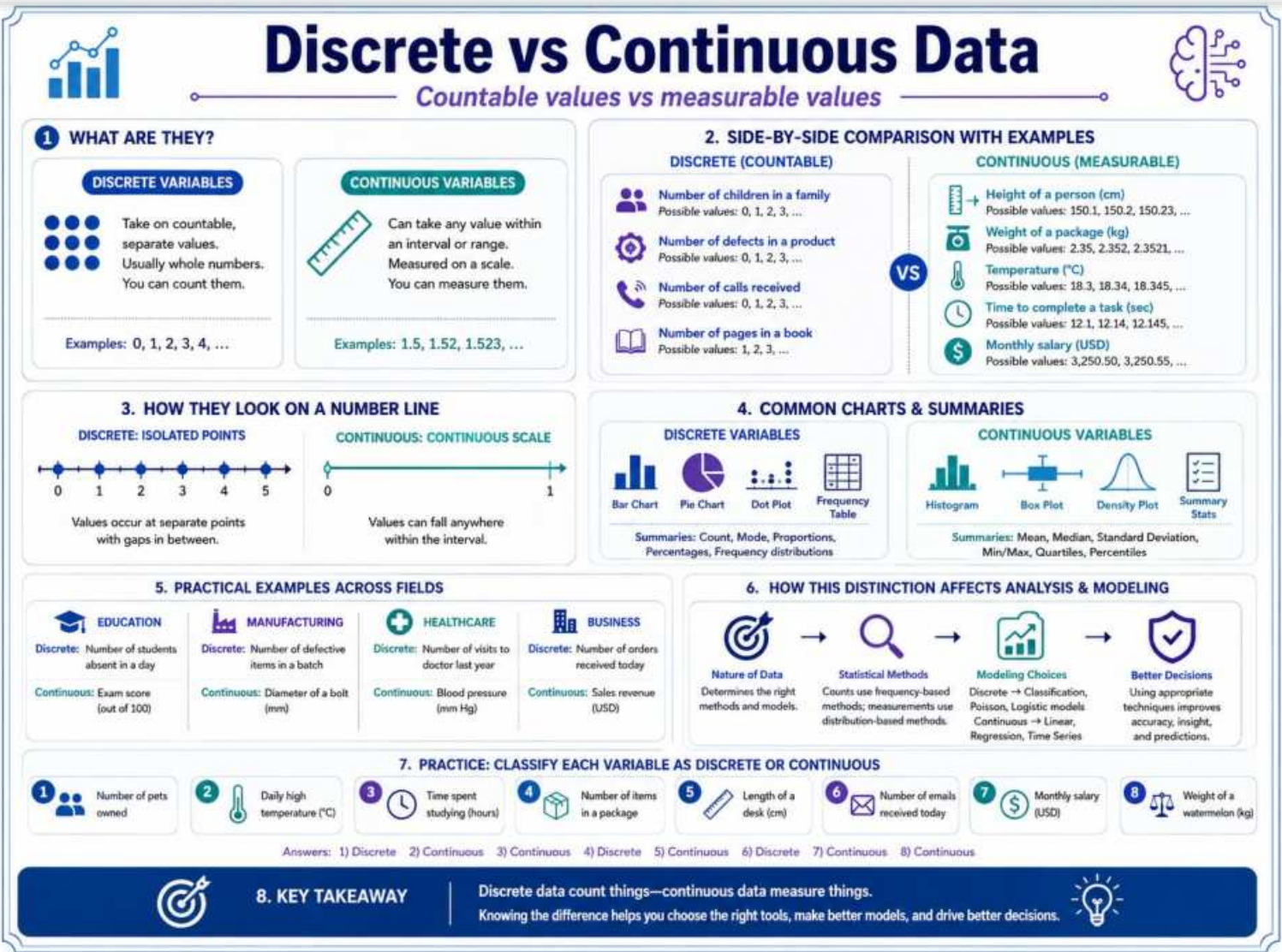
Types of Data: Qualitative and Quantitative Variables

Module 2: Types of Data and Measurement



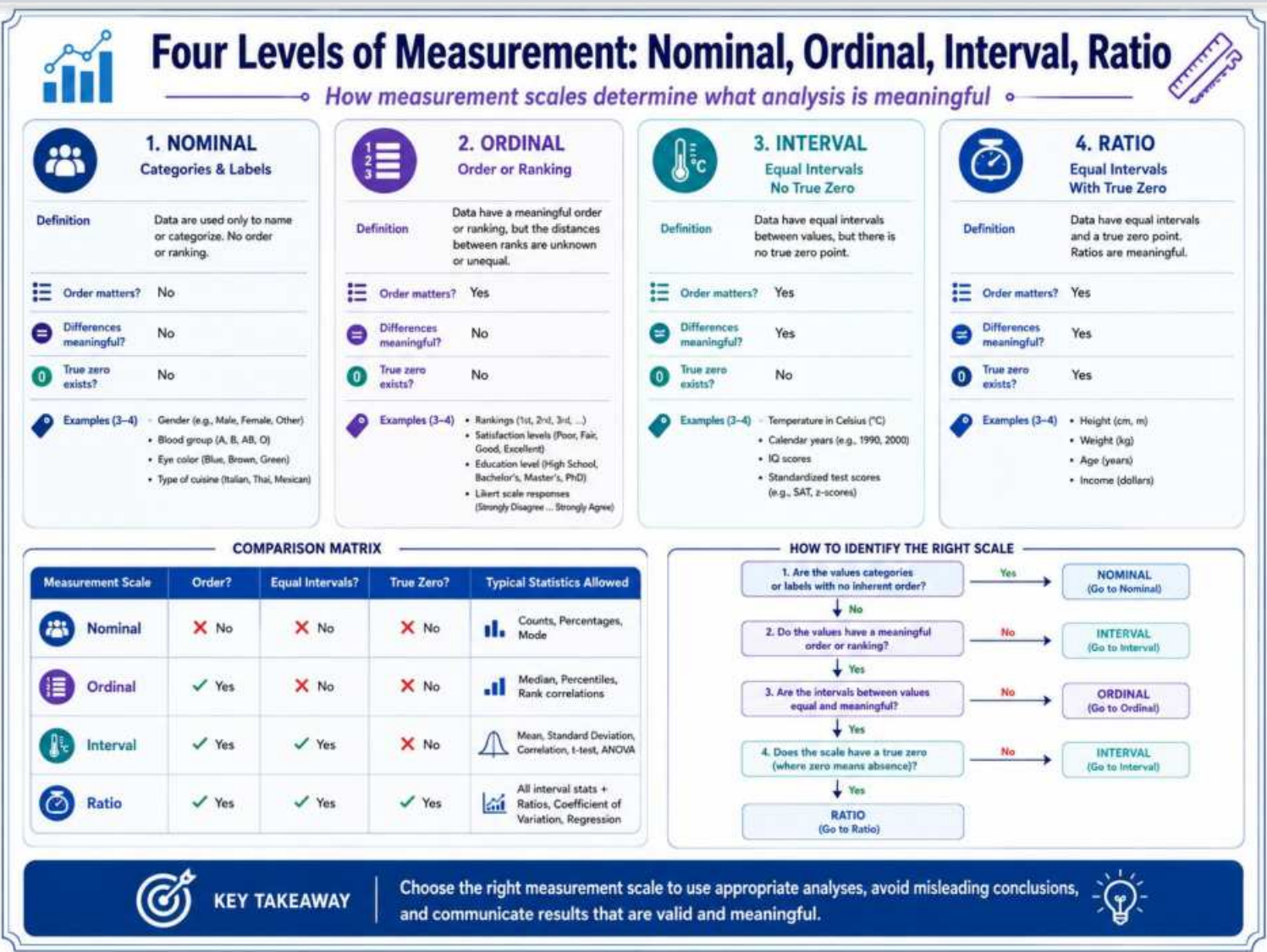
Discrete vs Continuous Data

Module 2: Types of Data and Measurement



Four Levels of Measurement: Nominal, Ordinal, Interval, Ratio

Module 2: Types of Data and Measurement



COMPARISON MATRIX

Measurement Scale	Order?	Equal Intervals?	True Zero?	Typical Statistics Allowed
Nominal	✗ No	✗ No	✗ No	Counts, Percentages, Mode
Ordinal	✓ Yes	✗ No	✗ No	Median, Percentiles, Rank correlations
Interval	✓ Yes	✓ Yes	✗ No	Mean, Standard Deviation, Correlation, t-test, ANOVA
Ratio	✓ Yes	✓ Yes	✓ Yes	All interval stats + Ratios, Coefficient of Variation, Regression

HOW TO IDENTIFY THE RIGHT SCALE

```
graph TD
    Q1[1. Are the values categories or labels with no inherent order?] -- Yes --> N1[NOMINAL  
(Go to Nominal)]
    Q1 -- No --> Q2[2. Do the values have a meaningful order or ranking?]
    Q2 -- No --> I1[INTERVAL  
(Go to Interval)]
    Q2 -- Yes --> Q3[3. Are the intervals between values equal and meaningful?]
    Q3 -- No --> O1[ORDINAL  
(Go to Ordinal)]
    Q3 -- Yes --> Q4[4. Does the scale have a true zero (where zero means absence)?]
    Q4 -- No --> I2[INTERVAL  
(Go to Interval)]
    Q4 -- Yes --> R1[RATIO  
(Go to Ratio)]
```

KEY TAKEAWAY

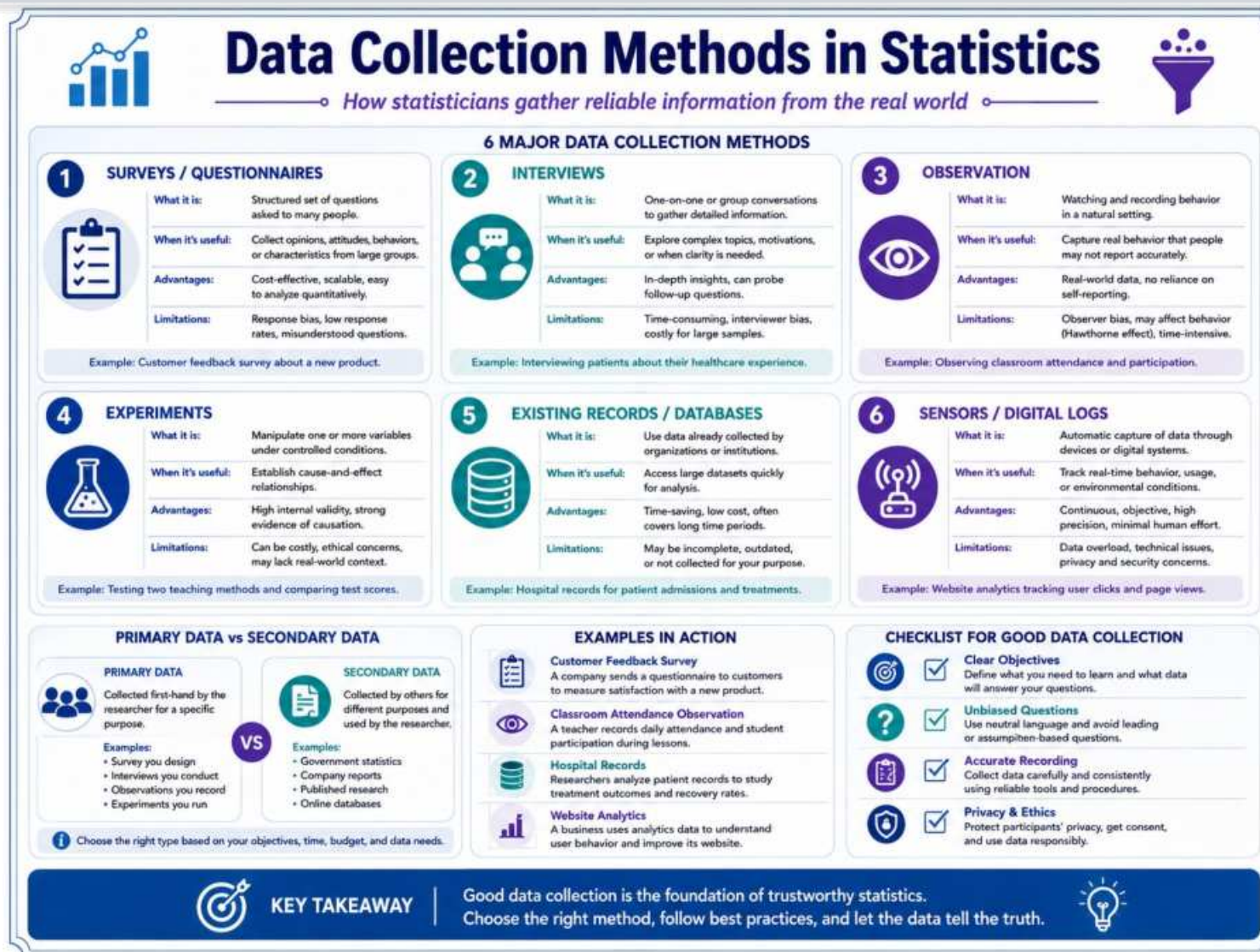
Choose the right measurement scale to use appropriate analyses, avoid misleading conclusions, and communicate results that are valid and meaningful.

Module 3: Data Collection and Sampling

4 infographic slides

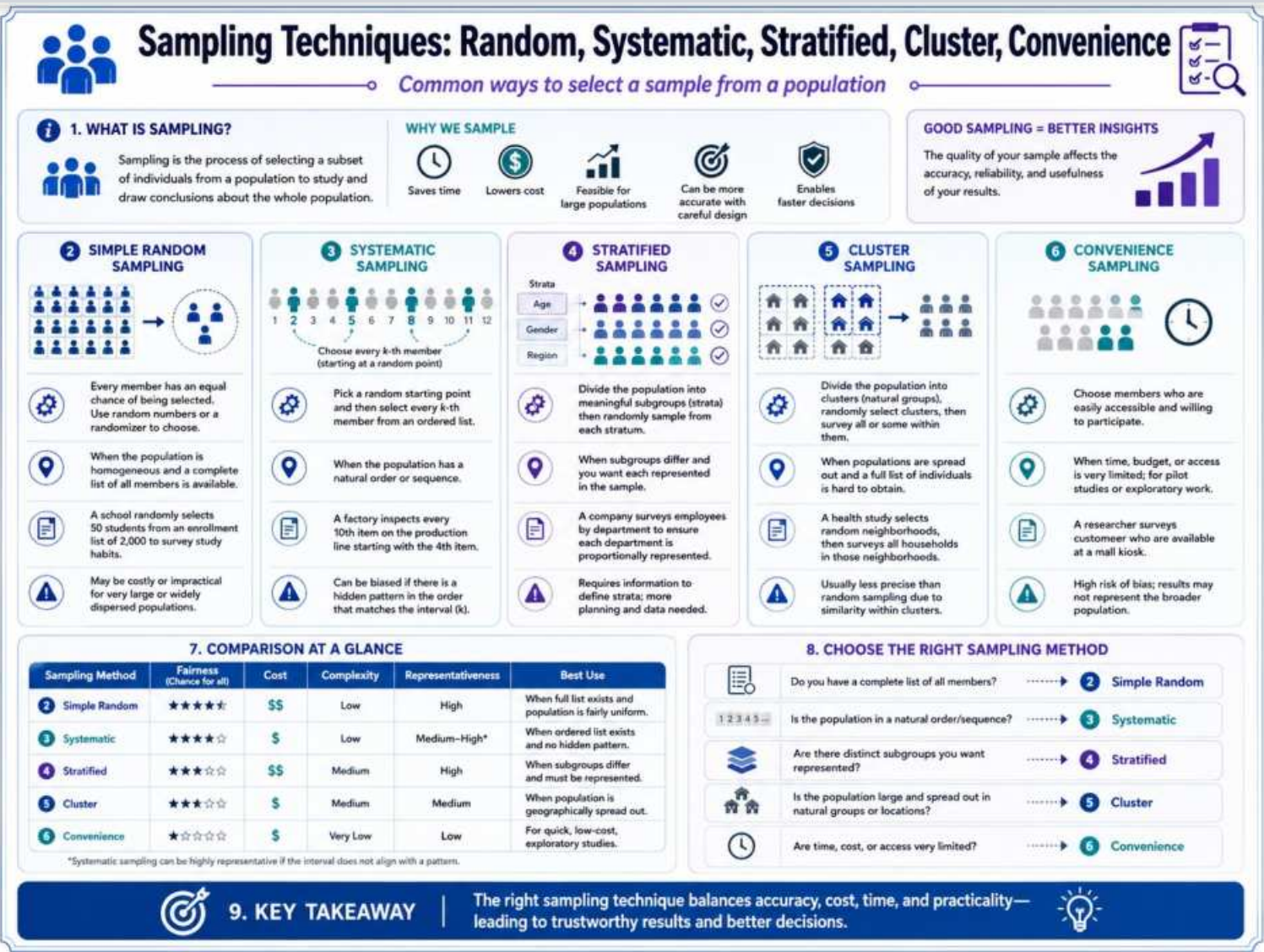
Data Collection Methods in Statistics

Module 3: Data Collection and Sampling



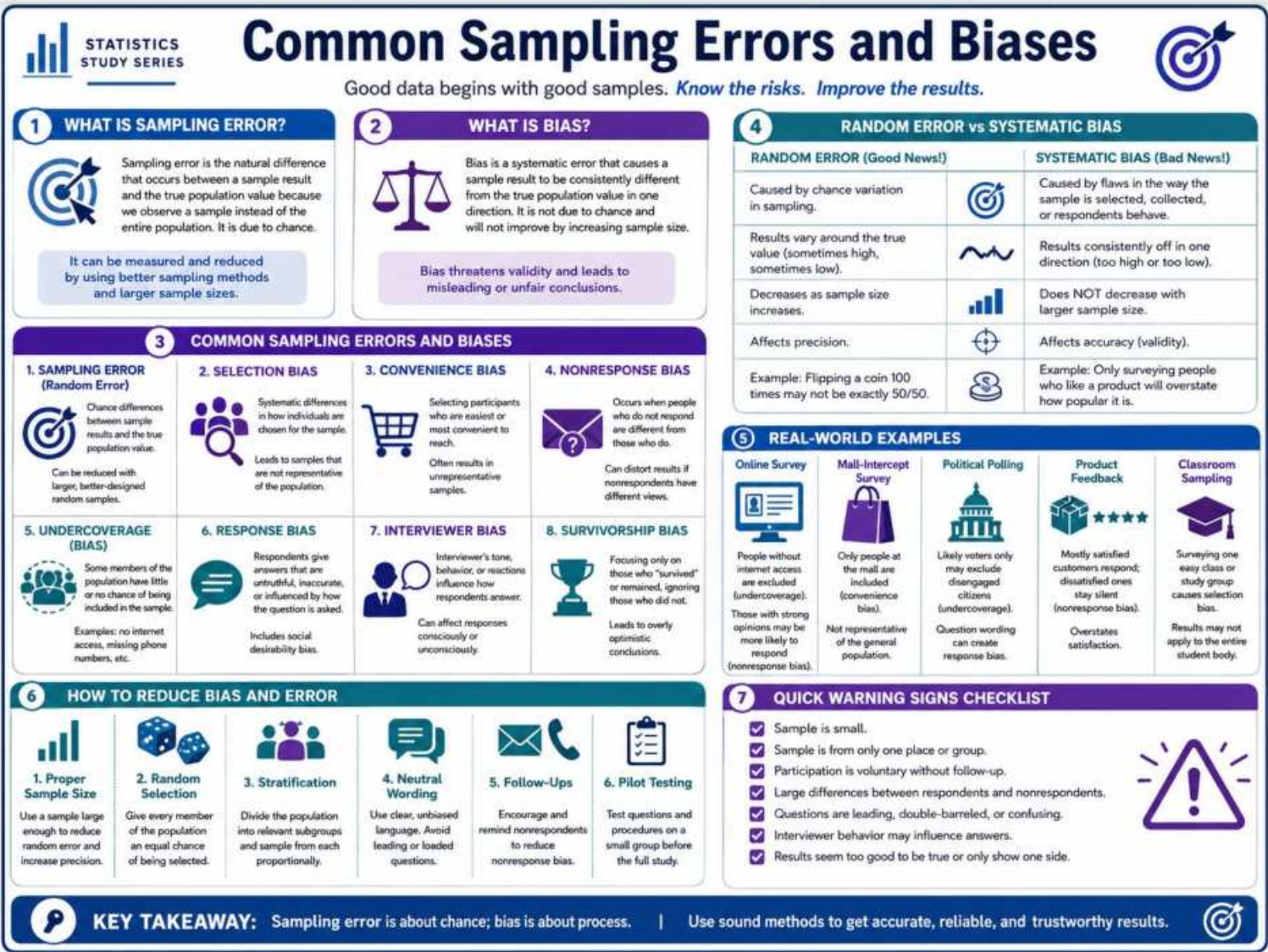
Sampling Techniques: Random, Systematic, Stratified, Cluster, Convenience

Module 3: Data Collection and Sampling



Common Sampling Errors and Biases

Module 3: Data Collection and Sampling



Sampling Technique Practice: Identify the Sampling Method

Module 3: Data Collection and Sampling

Sampling Technique Practice: Identify the Sampling Method

PRACTICE • RECOGNIZE • REVISE • SUCCEED

1 RECAP: THE FIVE SAMPLING METHODS

1 Simple Random Sampling

Everyone has an equal chance of being selected.

2 Systematic Sampling

Select every k^{th} member after a random start.

3 Stratified Sampling

Divide into groups (strata) and sample from each group.

4 Cluster Sampling

Randomly select clusters, then survey all in those clusters.

5 Convenience Sampling

Choose whoever is easily available or convenient.

2 DECISION CLUES: HOW TO RECOGNIZE EACH METHOD

1 **Simple Random:** Based on chance; no pattern; equal opportunity.

2 **Systematic:** Look for "every k^{th} " or a fixed interval.

3 **Stratified:** Population is divided into meaningful subgroups (strata).

4 **Cluster:** Groups (clusters) are selected; often entire clusters are used.

5 **Convenience:** Selected due to ease of access or availability.

3 PRACTICE: IDENTIFY THE SAMPLING METHOD

Read each scenario and identify the sampling method used.

1 A school has a list of all 600 students. Using a random number generator, 50 students are selected.

Answer: Simple Random Sampling

2 At a store, the first customer is chosen randomly. After that, every 10th customer is surveyed.

Answer: Systematic Sampling

3 Employees are divided by department (HR, Sales, Finance). A random sample is taken from each.

Answer: Stratified Sampling

4 Two neighborhoods are randomly chosen. Every household in those neighborhoods is surveyed.

Answer: Cluster Sampling

5 A survey is conducted by asking whoever is nearby in the mall to participate.

Answer: Convenience Sampling

6 From a jar containing employee ID slips, 75 slips are mixed and 25 are drawn at random.

Answer: Simple Random Sampling

7 Starting with the 5th name on a list, every 15th name is selected until the sample size is met.

Answer: Systematic Sampling

8 The city is divided into age groups (18–25, 26–40, 41–60, 60+). A sample is taken from each group.

Answer: Stratified Sampling

9 Three schools are randomly selected from the district. All students in those schools are surveyed.

Answer: Cluster Sampling

10 Patients are surveyed as they leave a clinic because they are easy to find and willing to respond.

Answer: Convenience Sampling

11 A list of 1,000 households is used. 200 households are selected randomly using a computer.

Answer: Simple Random Sampling

12 Every 10th item is picked from an inventory list after a random start.

Answer: Systematic Sampling

4 MINI CHALLENGE: WHAT'S THE METHOD?

Identify the sampling method for each scenario.

A A company divides its branches by region. From each region, 5 branches are randomly selected. All customers in those branches are surveyed.

Answer: _____

B In a factory, starting with the 7th product off the line, every 25th product is inspected.

Answer: _____

C Students who are waiting outside the library are asked to fill out a short feedback form.

Answer: _____

COMMON MISTAKES TO AVOID

- Confusing systematic with stratified - Systematic uses a fixed interval, stratified uses groups (e.g., by age, department).
- Confusing cluster with stratified - Cluster selects whole groups (e.g., schools, neighborhoods), stratified samples within each group.
- Assuming convenience is random - Convenience is based on ease, not chance.
- Overlooking the random start in systematic sampling.
- Forgetting that in cluster sampling, all (or many) elements in the selected clusters are included.

ANSWER KEY

Practice (3):

1: Simple Random	2: Systematic	3: Stratified
4: Cluster	5: Convenience	6: Simple Random
7: Systematic	8: Stratified	9: Cluster
10: Convenience	11: Simple Random	12: Systematic

Mini Challenge (4):

A: Cluster	B: Systematic	C: Convenience
------------	---------------	----------------

(Check your reasoning using the decision clues above!)

KEY TAKEAWAY: Look for patterns, groups, clustering, or ease of access. Use the decision clues to match each scenario to the correct sampling method!

Infographic 12 of 15

Module 4: Study Design and Application

3 infographic slides

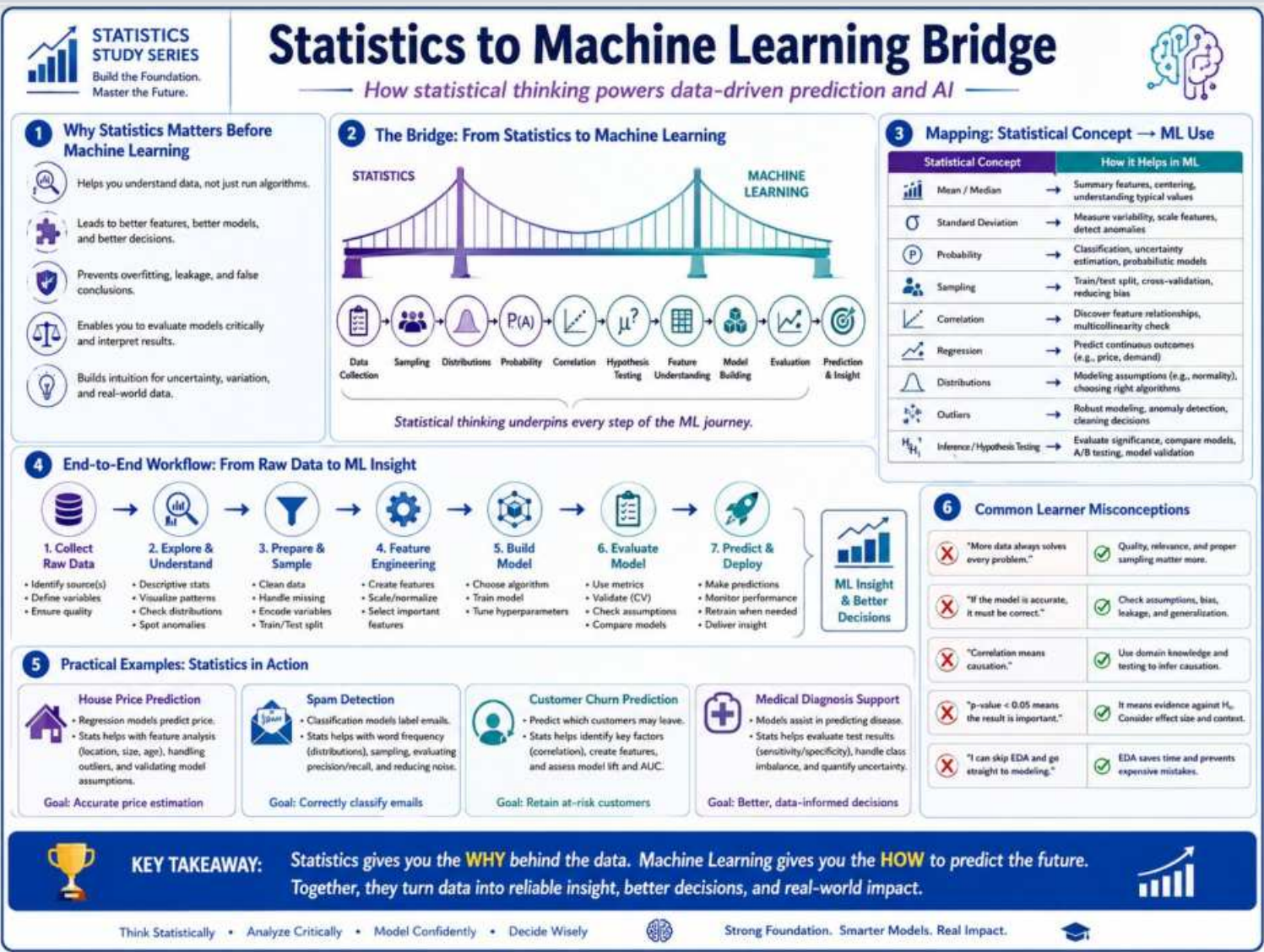
Observational vs Experimental Studies

Module 4: Study Design and Application



Statistics to Machine Learning Bridge

Module 4: Study Design and Application



Choosing the Right Statistical Method: A Beginner Decision Map

Module 4: Study Design and Application

